NAME: ALLAN NJUGUNA NGUGI

EMAIL: ngugiallan05@gmail.com

ADMISSION NUMBER: CS-SA12-26007

Table of Contents

Introduction 3

Task 2: Hello World 3

Task 3: Mathematical Operators 4

Task 4: Variables and Data Types 7

Task 5: Logical and Boolean Operators 8

Task 6: Introduction to If Statements 8

Task 7: Loops 10

Task 8: Introduction to Functions 11

Task 9: Files 13

Task 10: Imports 15

Conclusion 16

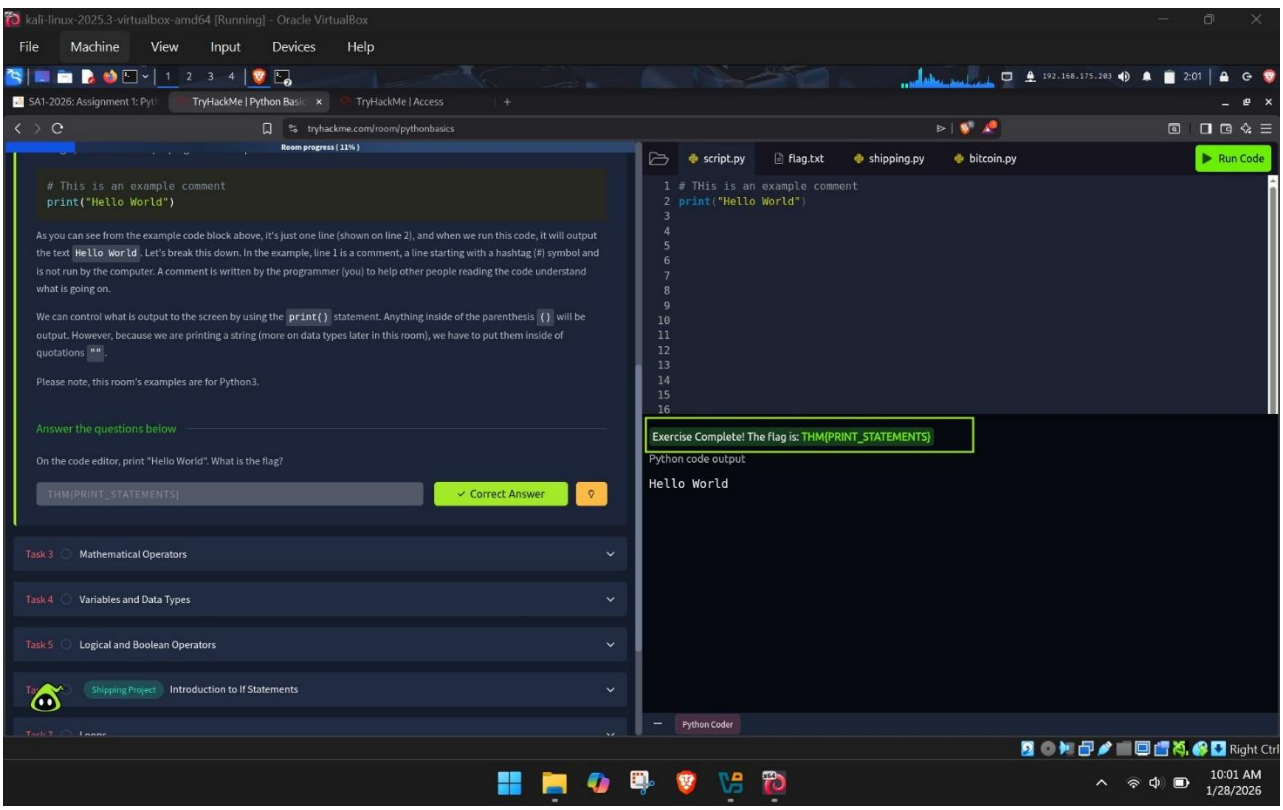
Link to Completed Module 16

Introduction

The room provided an introduction to Python basics which demonstrate how scripting functions as a security force multiplier for developing custom hacking tools and analysis tools. The reason for programming teaches essential programming skills through its content which includes syntax and variables and data structures while one practices their skills in a browser-based editor to complete challenges. The material first teaches fundamental coding logic before it develops one's ability to automate both defensive and offensive operations.

Task 2: Hello World

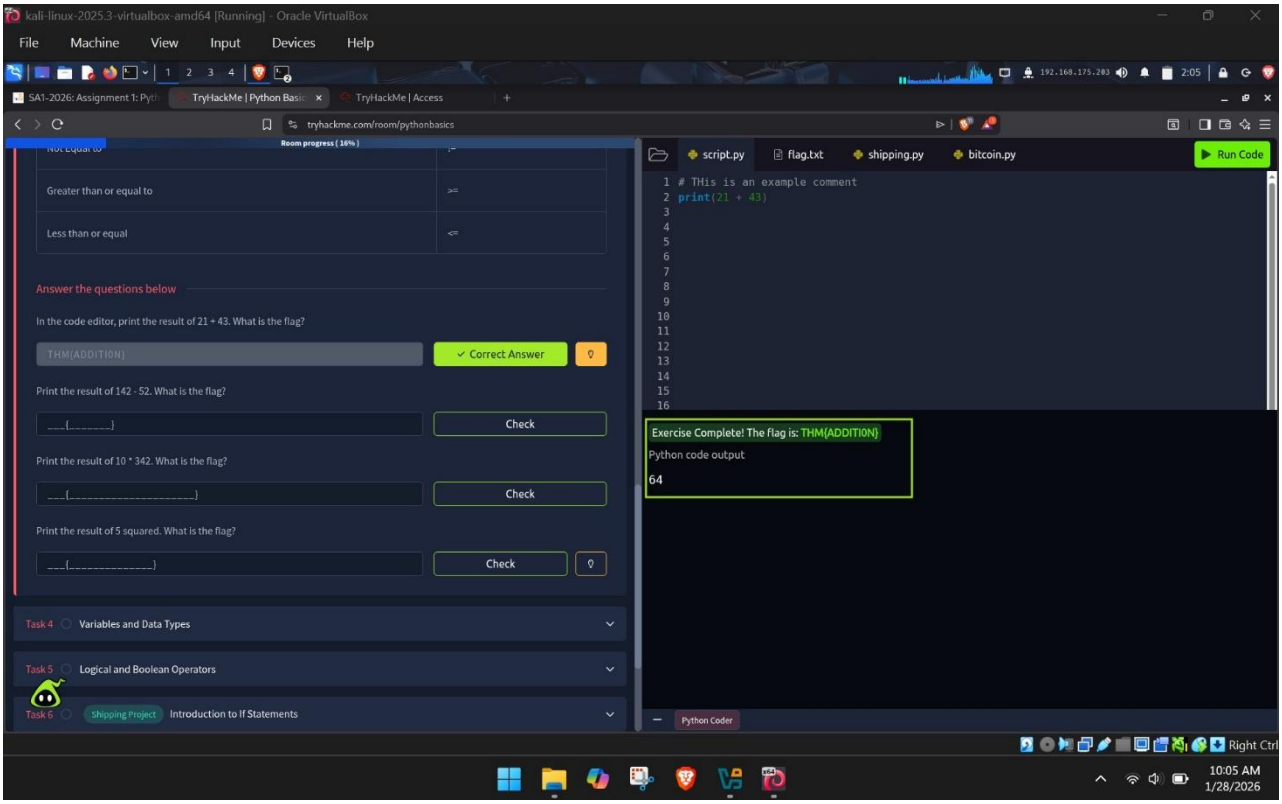
This task simply required one to write a simple python program to output Hello World and obtain the flag required to answer the question.



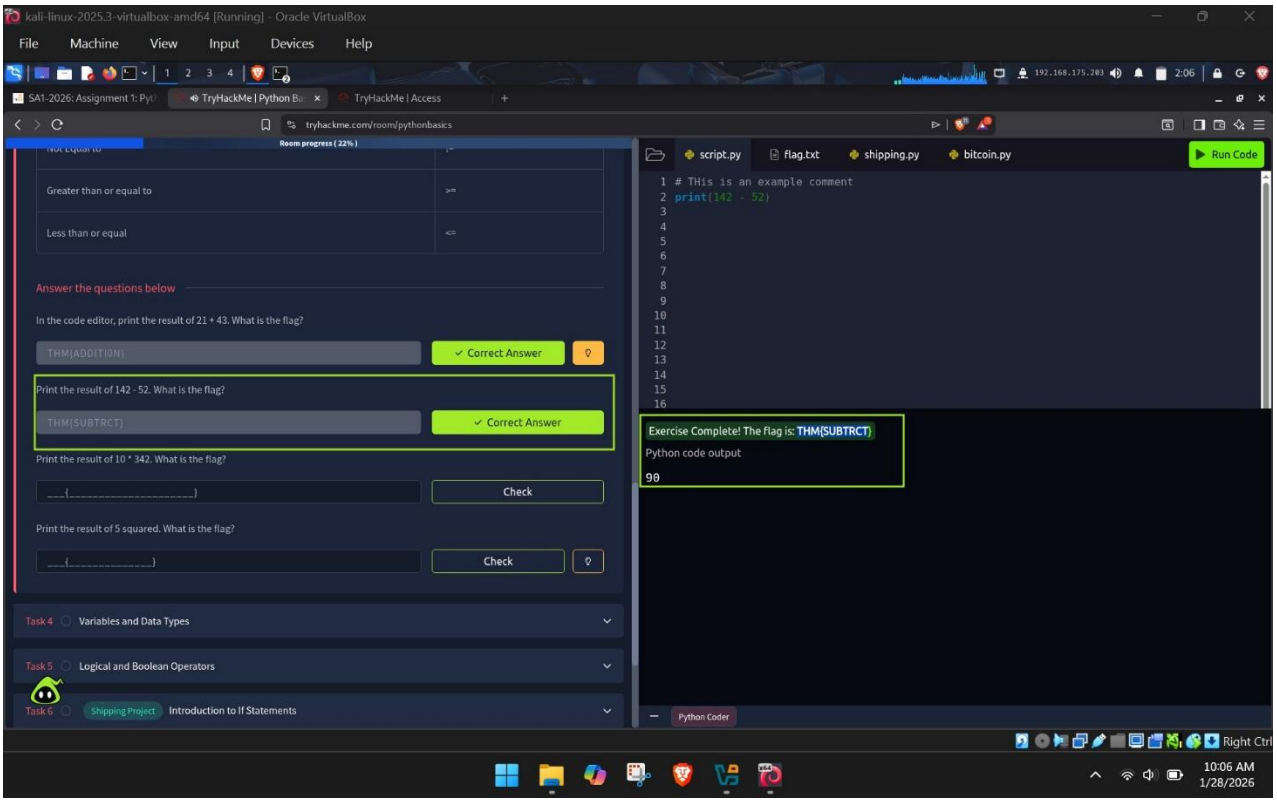
Task 3: Mathematical Operators

This task involved the use of operators to perform math arithmetic. The operators included operations such as addition, subtraction, multiplication as well as division. Other parameters that were involved included comparison such as the greater than, the less than amongst others.

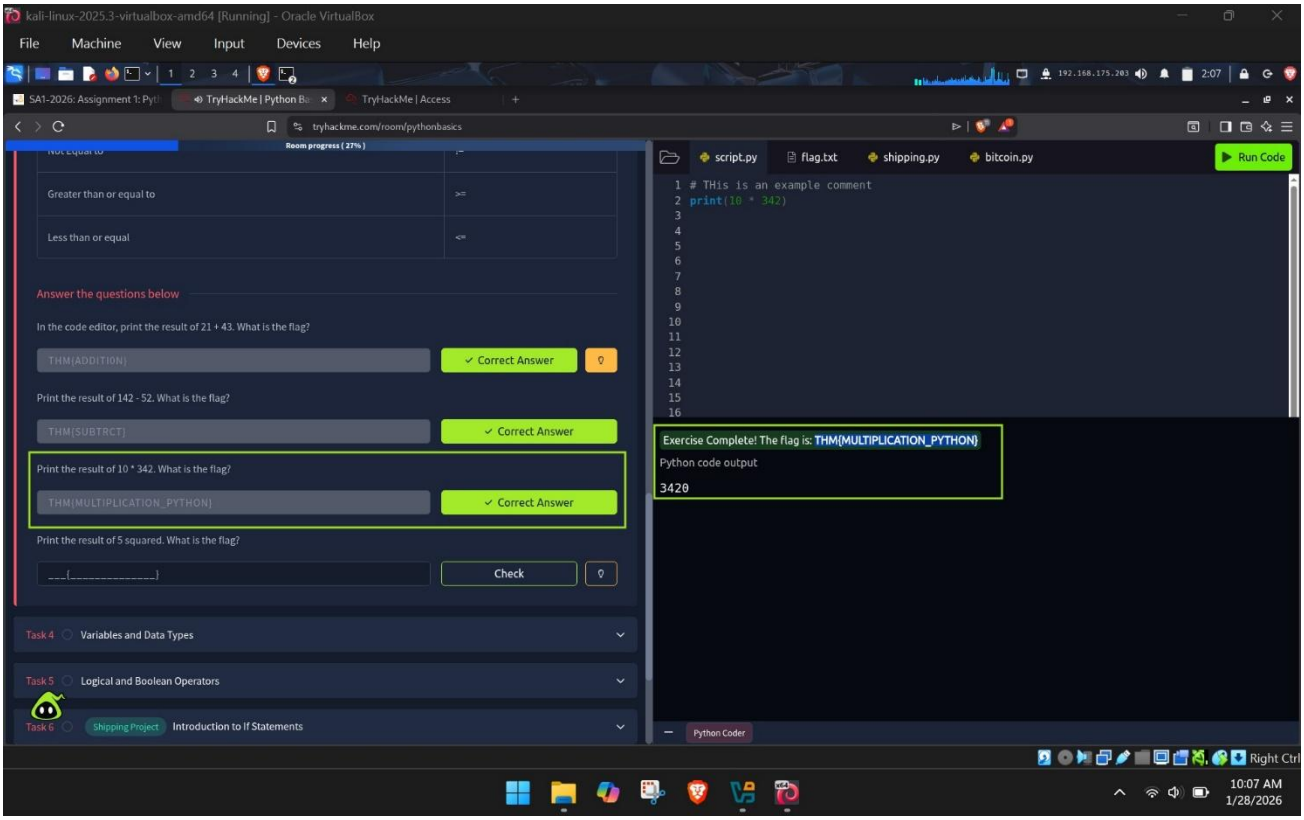
The questions covered are shown below:



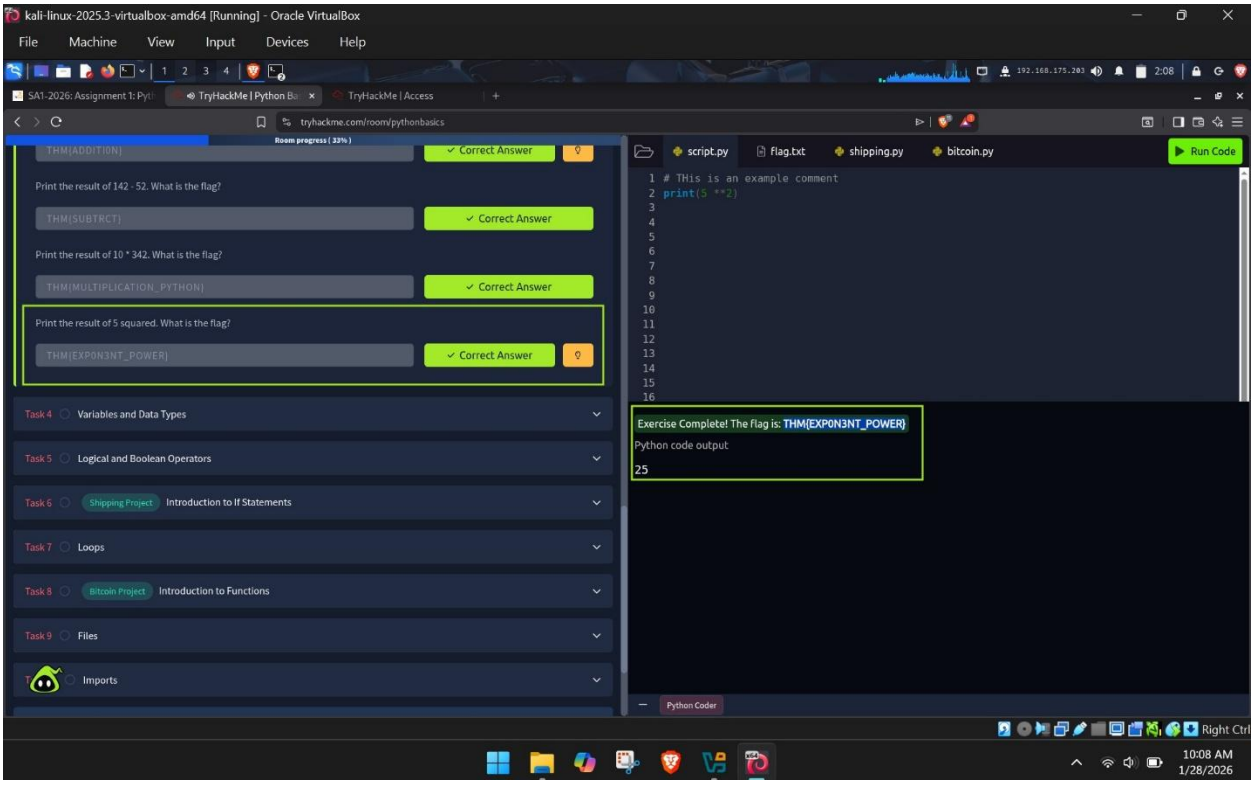
The question prompted the use of the additional operator to add 21 and 43 in the python editor, of which the result is 64, and the flag obtained was used to answer the question as shown.



The question prompted the use of the subtraction operator to subtract 52 from 142, with the result being 90, and the flag was obtained to answer the question as shown.



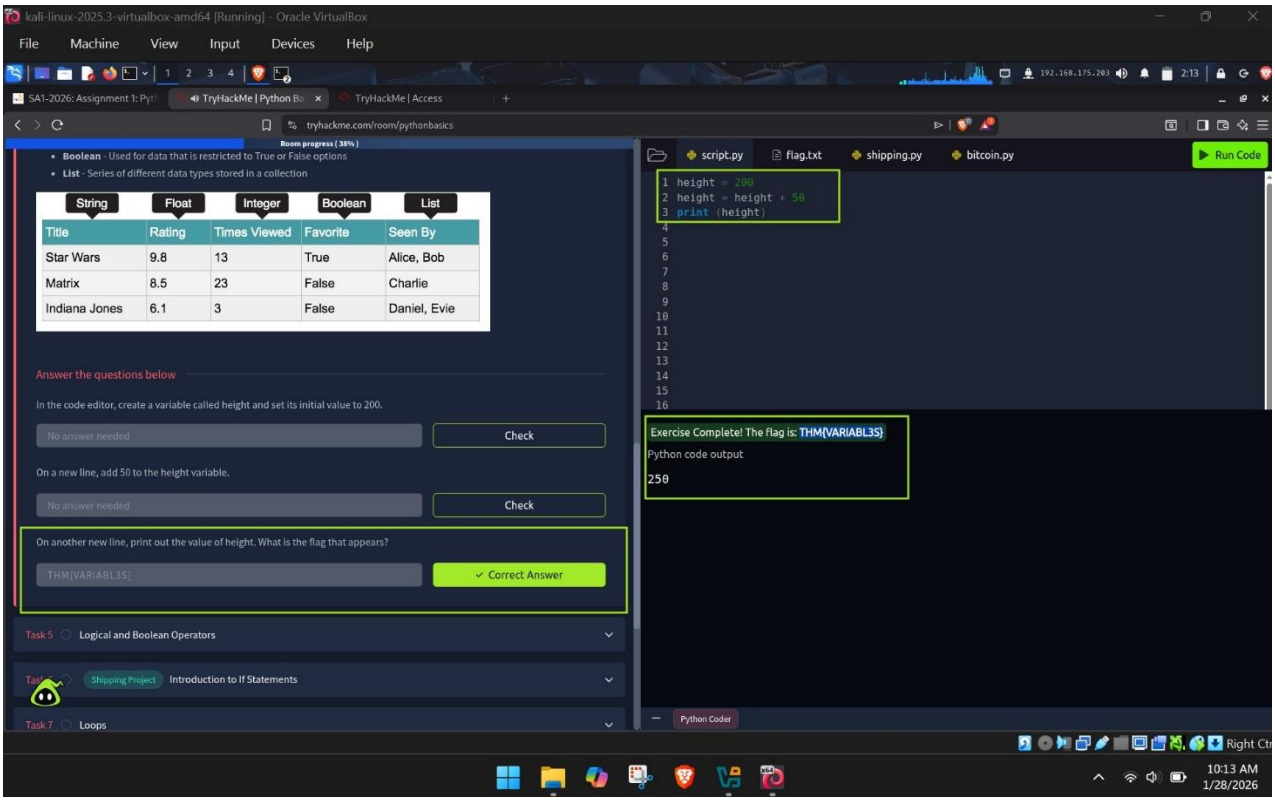
The question prompted the multiplication operation of 10 and 342, with the result being 3420, and the flag obtained was used to answer the question as shown.



The question prompted the exponent operator; with the number being 5, and I squared it to find 25 as the answer, with the flag obtained to answer the question as shown.

Task 4: Variables and Data Types

The task touched on variables that allow one to store and update data in a computer program, as well as different datatypes such as a string, integer and boolean amongst others.



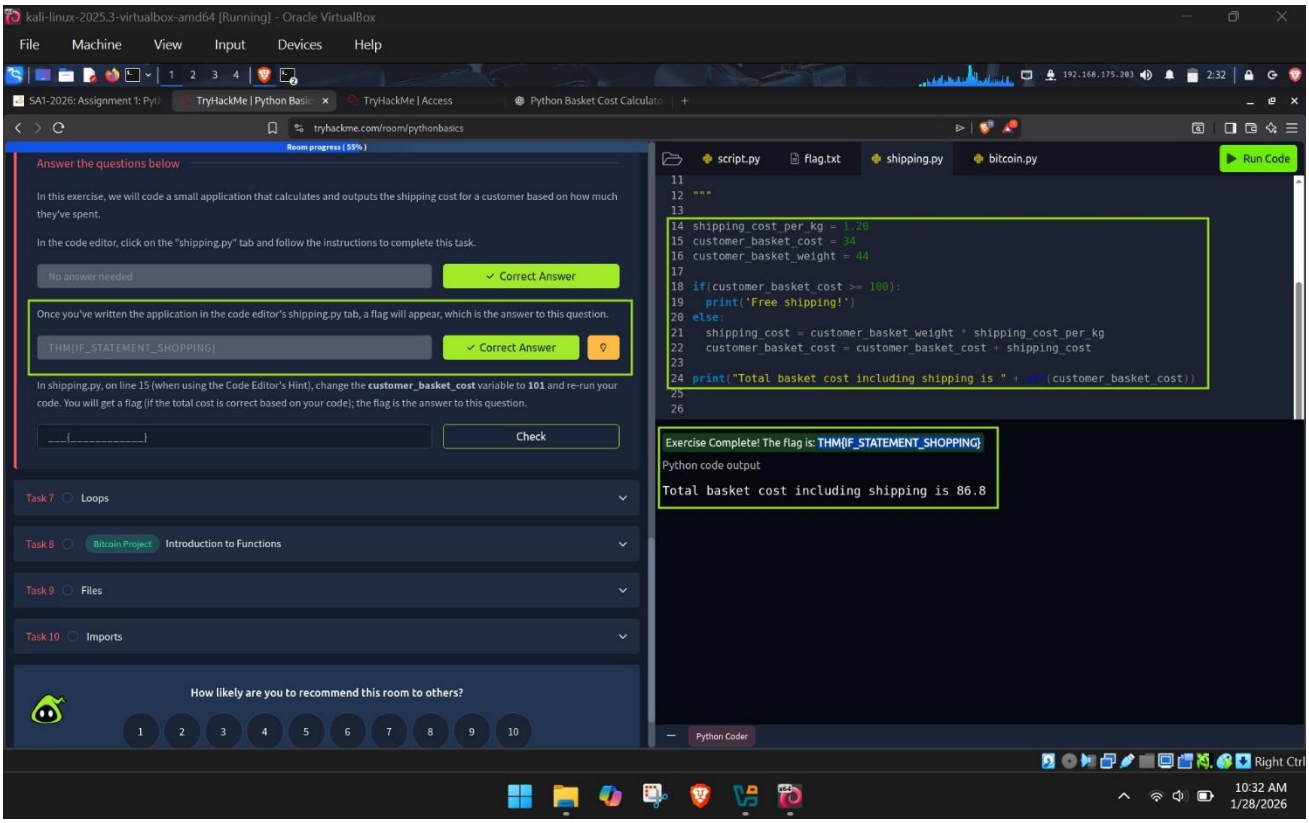
The question required one to create a variable with the initial value being 200. Then, the same variable was used but with an additional value added to the initial value and the result printed in the terminal. The output is shown above.

Task 5: Logical and Boolean Operators

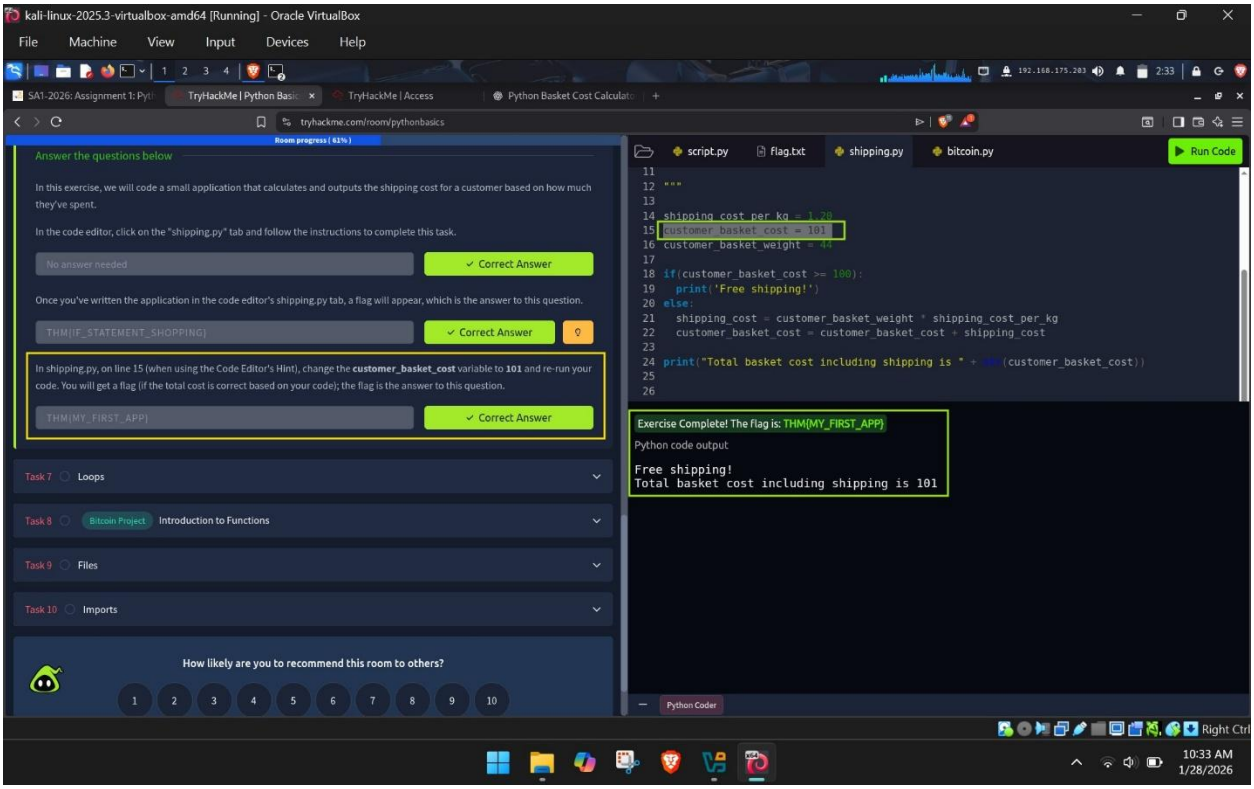
This task focused on the various logical operators such as greater than, less than, greater or equal to, less or equal to and equivalence. It also touched on the Boolean operators, AND, OR and NOT.

Task 6: Introduction to If Statements

The task touched on the IF statement, which allow a program to make decisions, based on certain conditions.



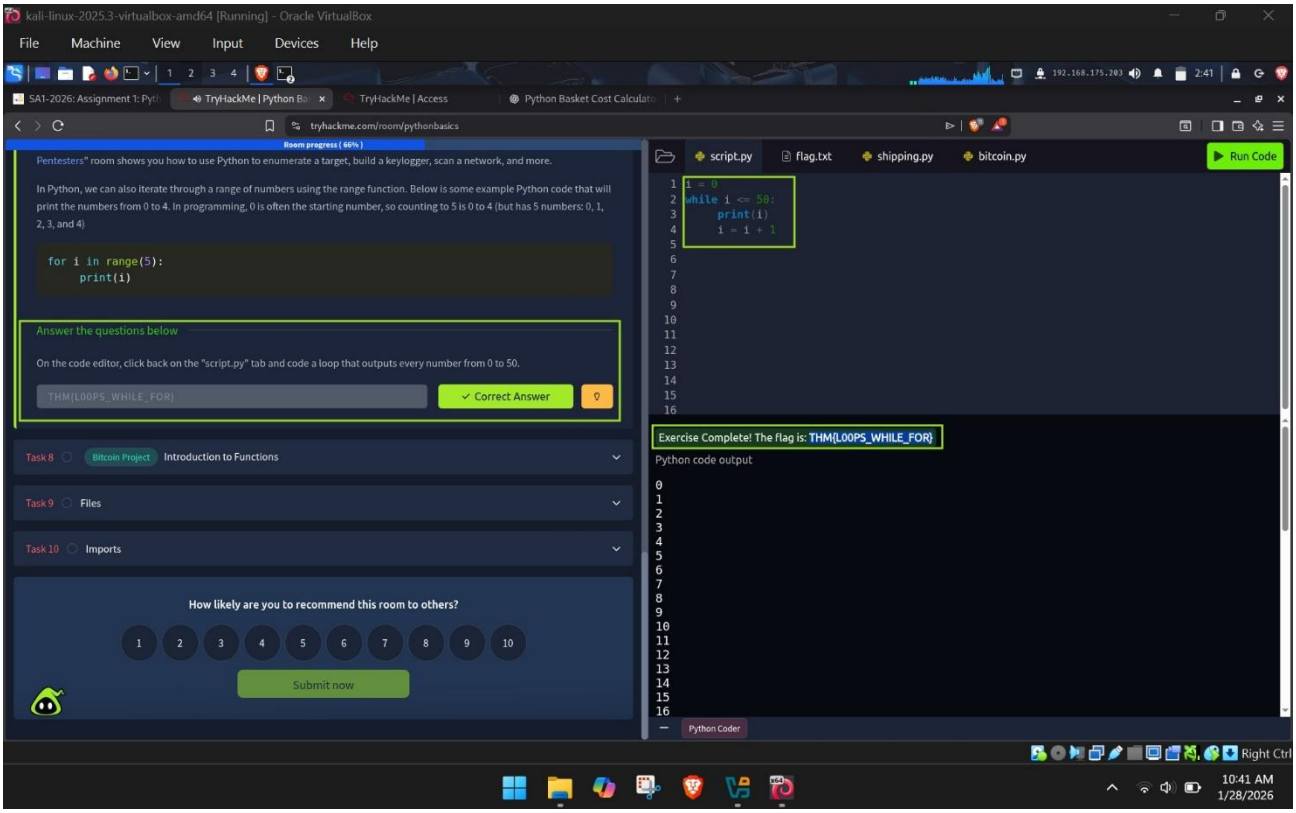
The task involved the calculation of the total cost of a client based on how much they have spent. If a customer spends more than $100, they get free shipping, otherwise, they are charged some shipping fee. The shipping cost would be calculated by multiplying the customer basket weight by the shipping cost per kg (1.20). Then the total cost would be calculated by adding the cost of the customer basket and the shipping cost. The program is shown above. In this case, the total basket cost for this client was 86.8 including shipping.



In this case, the basket cost for this client was more than 100, meaning that the customer was not charged any shipping fee.

Task 7: Loops

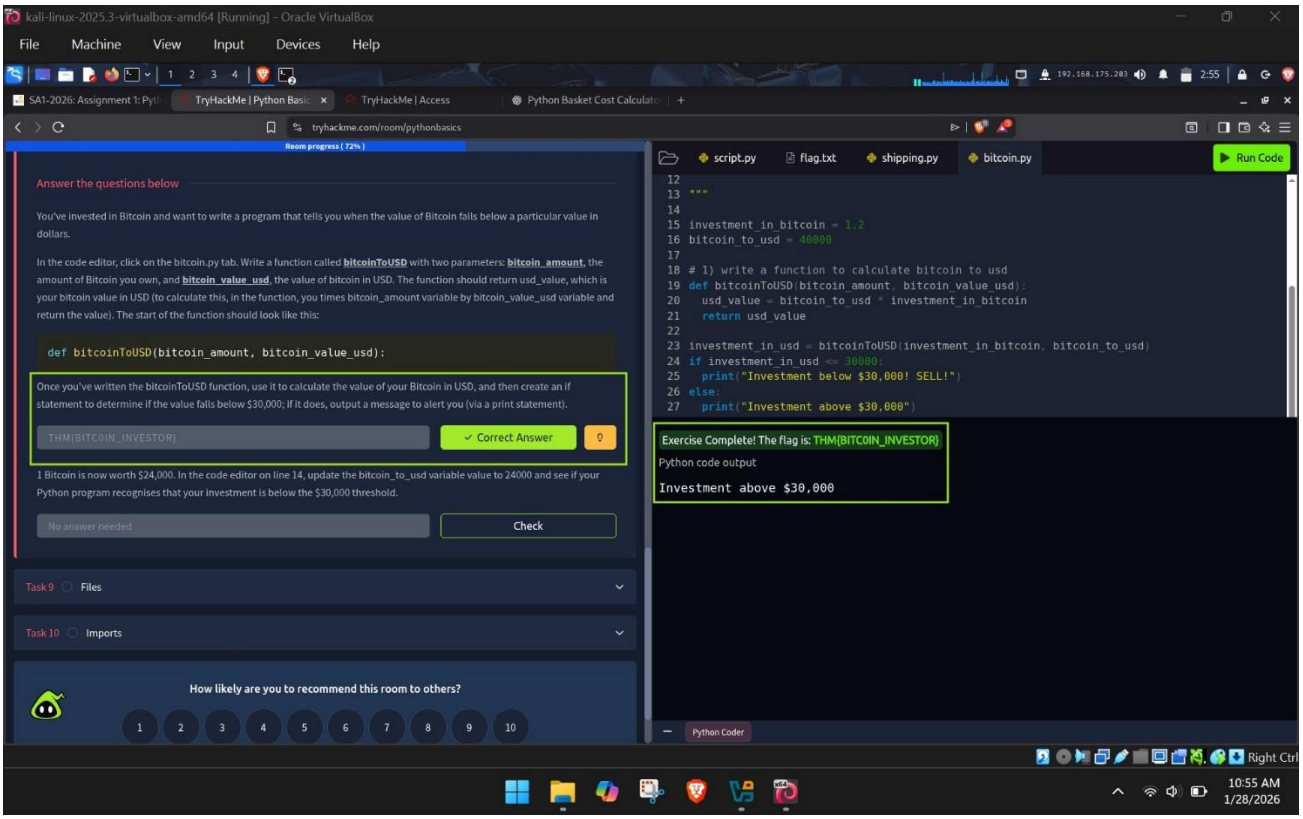
This task covered the for and while loops; with the for loops used to iterate over a sequence such as a list, and the while loop for determining hoe many times the loop should run based on a condition.



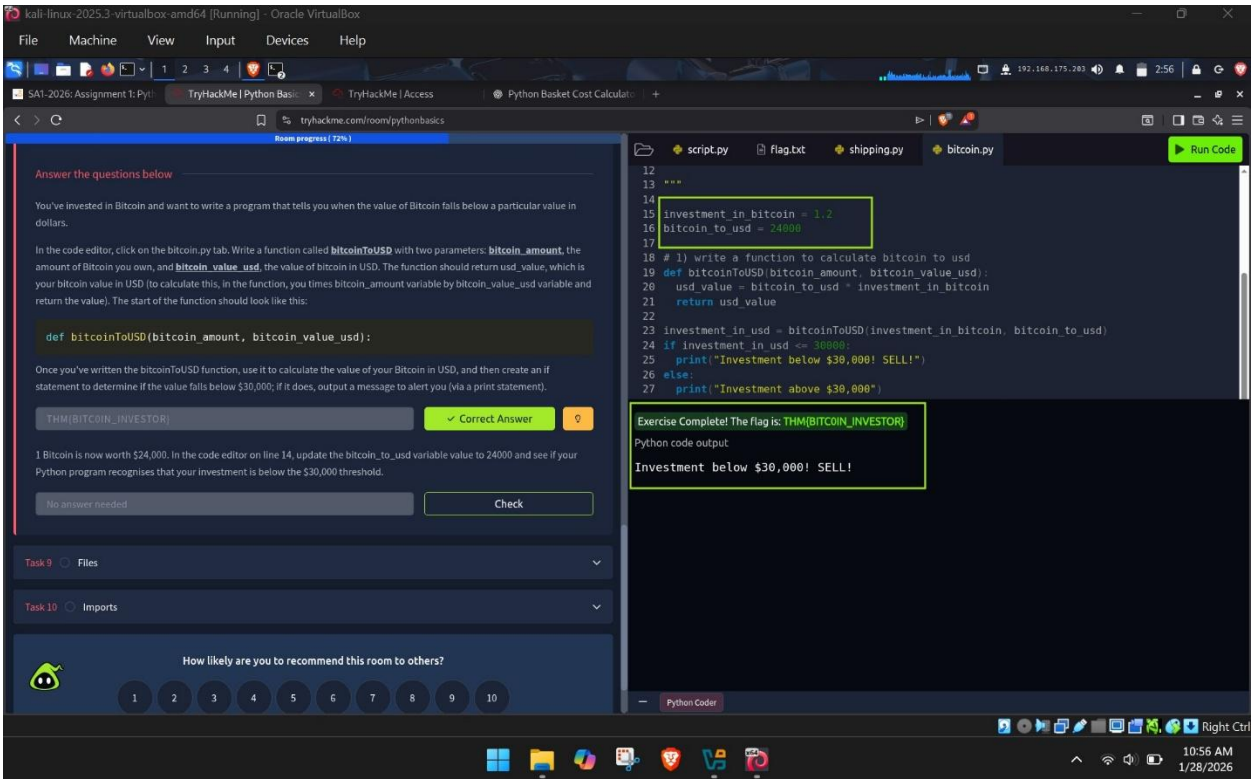
The question prompted a while loop that would run a loop that would output every number from 0 to 50. The initial variable ‘i’ set to 0 and the while loop to set the condition of printing the output so long as the variable is less than or equal to 50.

Task 8: Introduction to Functions

The task focused on functions, which are very essential in eliminating repetitive code.



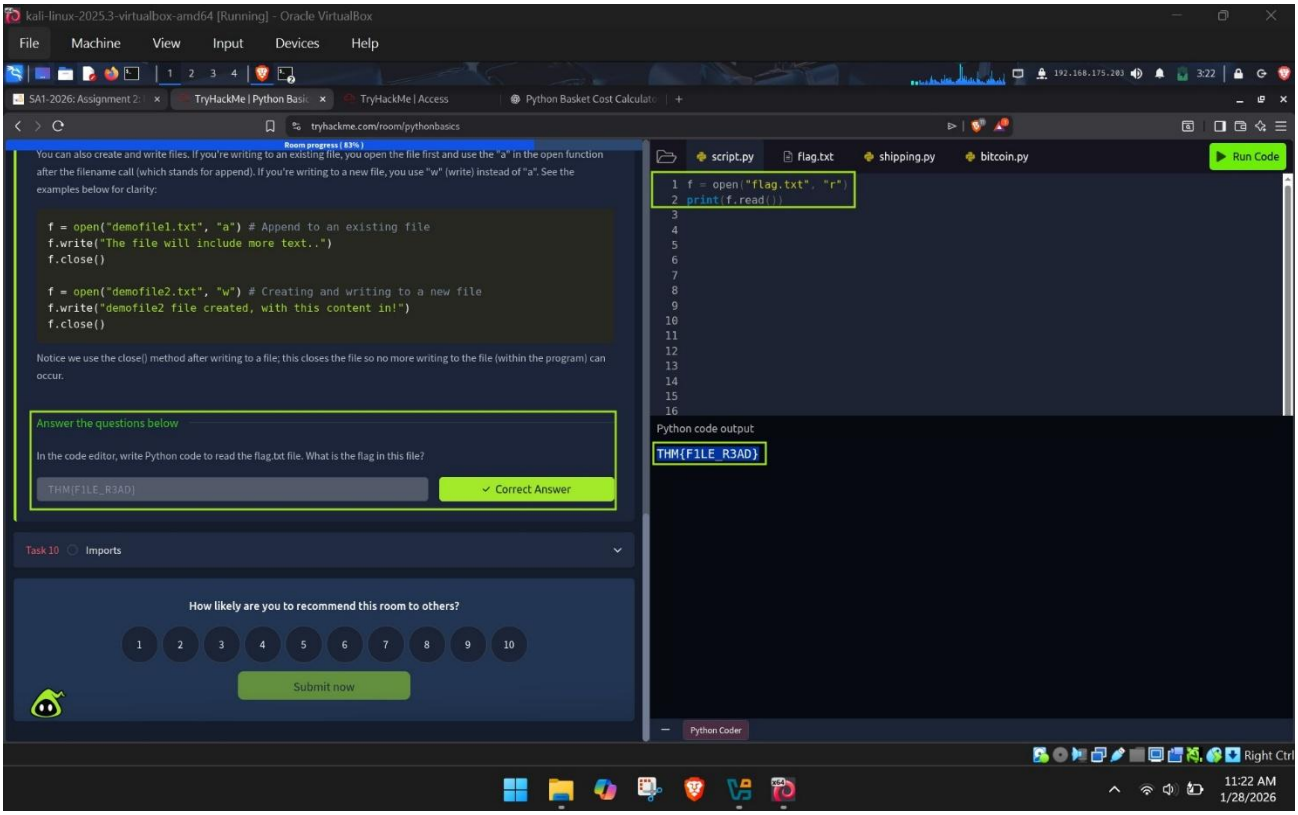
The program involved using functions to tell a user who invented in bitcoin on whether the value of bitcoin has fallen below a particular threshold ($30,000 in this case). If it falls below this threshold, the user is advised to sell but if it it above 30, 000, the user need not sell. The value in usd is calculated by multiplying the bitcoin to usd by the investment in bitcoin.



In this case (above), the calculated value falls below the threshold ($30,000) which then the user is prompted to sell.

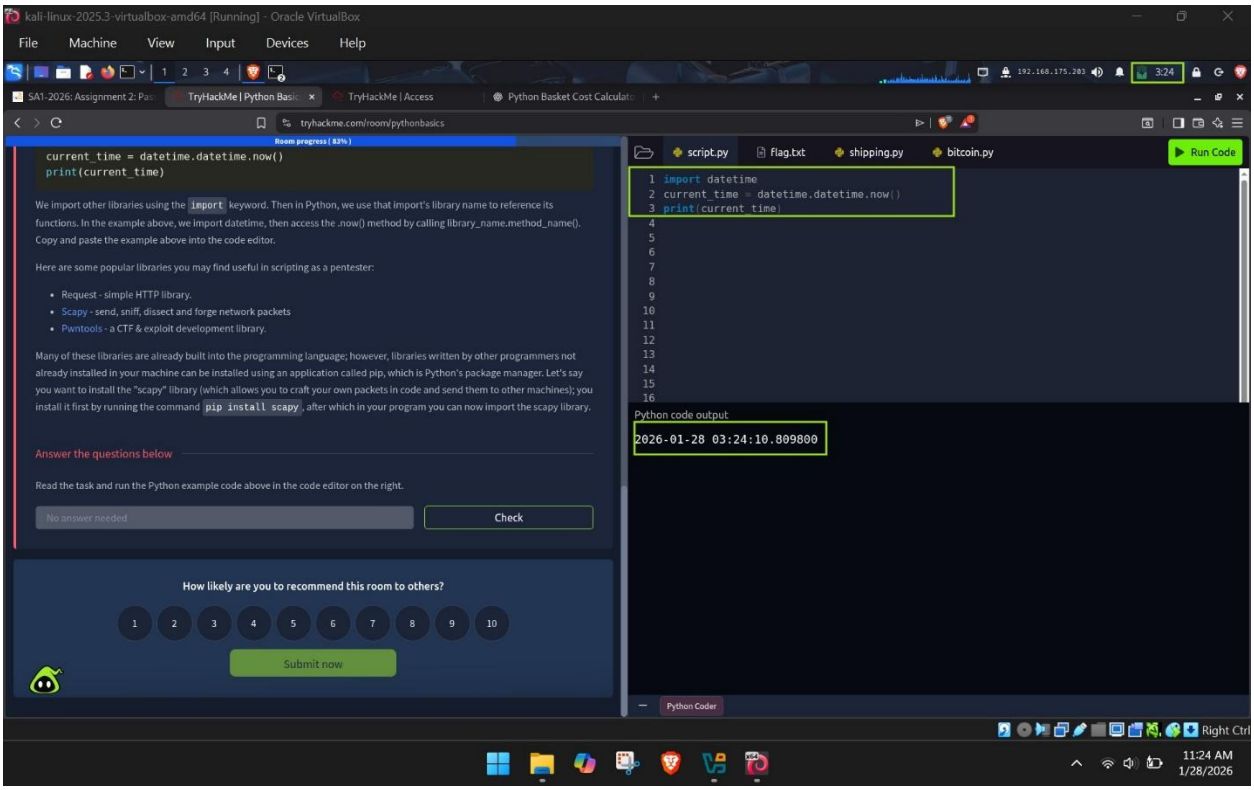
Task 9: Files

The task concentrates on the ability of reading and writing content in files through Python.



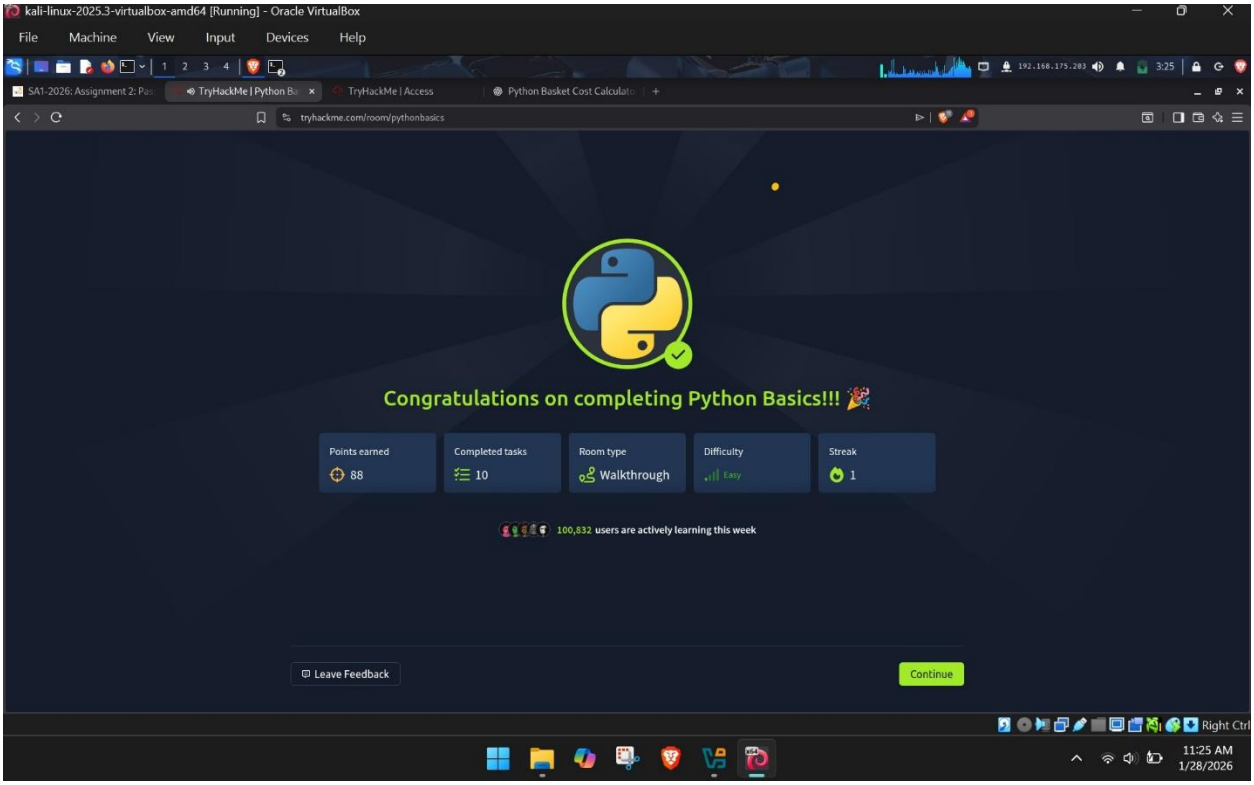
The process involved reading the contents of a flag.txt file to obtain the flag used to answer the question. The simple process involved just using the open function and using the r parameter for reading the contents.

Task 10: Imports



The task involved importing libraries such as Request and Scapy which are popular libraries.

Module Completion



Conclusion

This task gave me the necessary knowledge that I needed to learn python for scripting functions, and even laid a foundation for learning how to develop hacking and analysis tools. It was a fun and interactive module; and bridged the gap for me between theory and real world automation.

Link to Completed Module

https://tryhackme.com/p/ngugiallan05